

Hand Gesture Recognition with CNNs

Visual Computing Project

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Project Overview

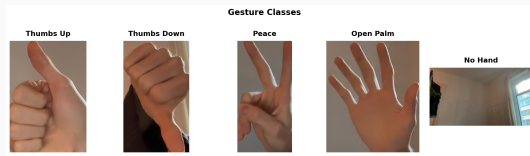
Goal: Classify hand gestures from webcam input.

5 Gesture Classes:

- Thumbs Up / Thumbs Down
- Peace Sign
- Open Palm
- No Hand (background)

Tech Stack:

- PyTorch (deep learning)
- OpenCV (image processing)
- MediaPipe (hand detection)



Sample Gestures

The Problem: Learning Backgrounds

Initial Approach:

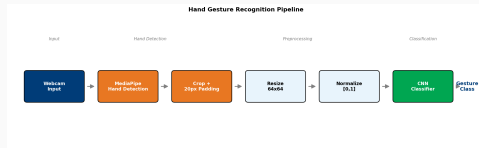
- Fed full webcam frames to CNN.

What Happened:

- Model learned **backgrounds**, not gestures.
- “Bookshelf = Thumbs Up”
- High training accuracy, poor generalization.

Root Cause:

- CNN found easier patterns (room features) than actual hand shapes.

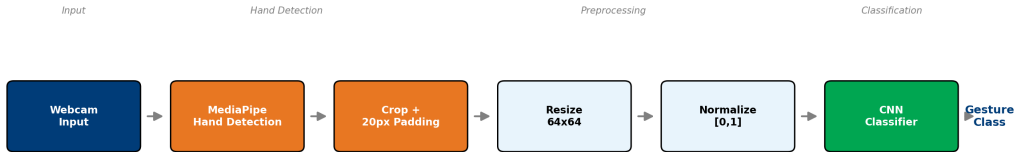


Solution:

Isolate the hand region first.

The Solution: MediaPipe Hand Cropping

Hand Gesture Recognition Pipeline



MediaPipe Hands

- Detects 21 hand landmarks (fingertips, knuckles, palm).

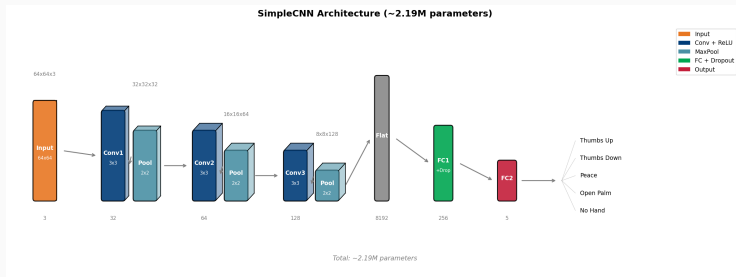
Bounding Box

- Extract hand region with 20px padding.

Result

- CNN only sees the hand, not the background.

CNN Architecture: SimpleCNN



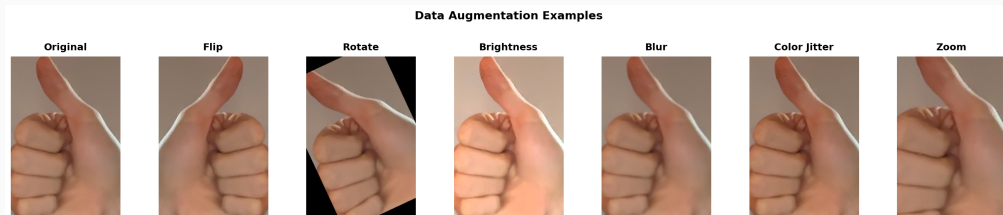
Structure:

- 3 Conv blocks (3 → 32 → 64 → 128)
- 2 FC layers (8192 → 256 → 5)
- Dropout (50%) for regularization

Details:

- Input: 64 × 64 RGB
- Total: ~ 2.19M parameters
- ReLU activations

Data Augmentation Strategy



6 Custom OpenCV Augmentations:

- Horizontal flip
- Rotation ($\pm 30^\circ$)
- Brightness ($0.5\times - 1.5\times$)
- Gaussian blur ($k = 3, 5, 7$)
- Color jitter (HSV)
- Zoom ($1.0\times - 1.3\times$)

Application:

- 50% probability per sample
- Training only (not test)
- No torchvision

Experimental Setup

Two Experiment Sets:

1. Augmentation Comparison:

- No augmentation (baseline)
- Single augmentations
- Flip + Rotate combined
- All augmentations

2. Dataset Size Impact:

- 50 / 100 / 200 / Full samples per class
- With and without augmentation

Training Configuration:

Parameter	Value
Optimizer	Adam
Learning rate	0.001
Epochs	20
Batch size	32
Split	80/20
Seed	42

~300 images per class
(collected via webcam)

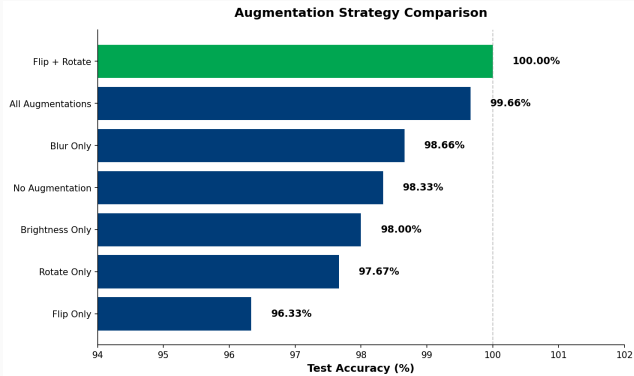
Results: Augmentation Comparison

Key Finding:

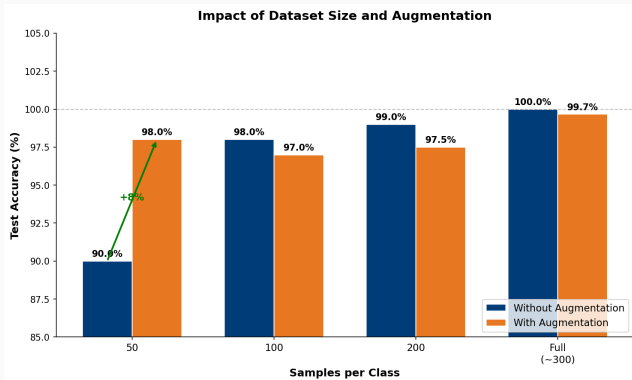
Flip + Rotate = 100%

Surprisingly, *all augmentations* (99.67%) performed **worse** than flip+rotate alone. **Why?**

- Geometric transforms match real gesture variation.
- Too many augmentations add noise.
- “More is not always better.”



Results: Dataset Size Impact

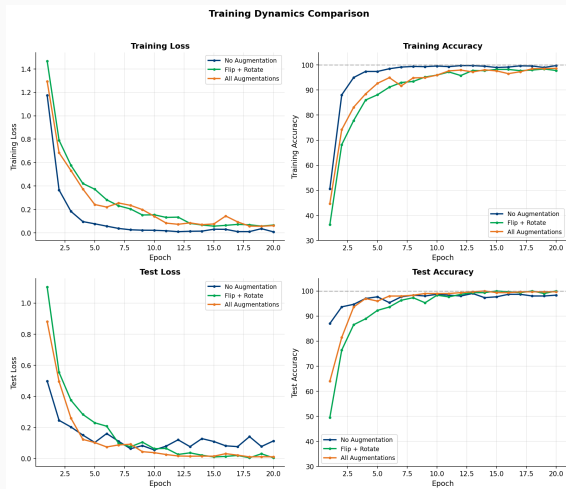


Key Finding: Augmentation helps most with **small data**.

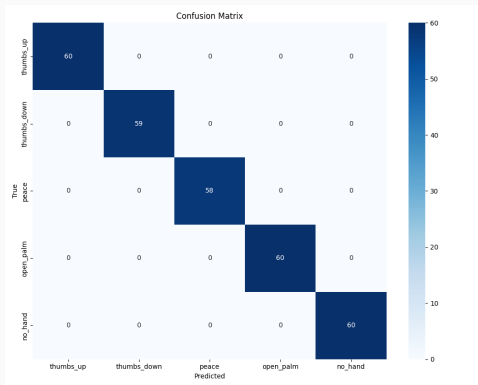
- **50 samples:** 90% → 98% (+8% improvement)
- **Full data:** Augmentation matters less.
- 100% possible without augmentation (if data is sufficient).

Practical Insight:
Augmentation is crucial when data is limited.

Training Dynamics



Final Model Performance



Confusion Matrix: Perfect Classification

Best Model: Flip + Rotate

- 100% test accuracy
- No misclassifications
- Real-time inference on webcam
- Robust across lighting conditions

Model saved as:

`models/model_flip_rotate.pth`

Key Takeaways & Limitations

What Worked:

1. **Hand cropping was crucial**
Isolate the signal from noise.
2. **Domain-relevant augmentation**
Flip + rotate match real variation.
3. **Simple architecture**
3 conv layers sufficient for 5 classes.

Counterintuitive: More augmentation $\neq$ better results.

Limitations:

- Only 5 gesture classes.
- Single hand detection.
- Trained on limited subjects (may not generalize to all hands).
- Requires good lighting for MediaPipe.

Future Work:

- More gesture classes.
- Multi-hand support.
- Diverse training data.

Live Demo

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